École Polytechnique Entrance Exam 2027

Filière Universitaire Internationale – FUI

Association des Polytechniciens Khmers (AXK)

May, 2026

Purpose of this document: This file summarizes the key information from the official École Polytechnique Entrance Exam FUI 2027 and organizes the Mathematics and Physics recommended background knowledge into a study plan with reference links.

Contents

1. Exam General Information

1.1 FUI Entrance Exam

The *Filière Universitaire Internationale (FUI)* is the international university entrance pathway for admission to the *Cycle Ingénieur* at *École Polytechnique.* It is intended for international students following a university-level science or engineering curriculum outside France. The selection process evaluates both the candidate's academic record and scientific ability through oral examinations.

Key Dates

The official calendar is subject to confirmation. Candidates should regularly check the official website for updates: official webpage.

Event	Date / Period
Opening of applications	Mid-May 2026
Application deadline	Mid-September 2026
Deadline for required documents	Mid-September 2026
Eligibility results	October 10, 2026, to be confirmed
Oral admission examinations	November 2 to 15, 2026
Admission results	November 26, 2026, to be confirmed
Academic entry	September 2027

Eligibility Requirements

Candidates must satisfy conditions related to nationality, age, language level, academic background, and physical fitness.

- Nationality: The FUI entrance exam is reserved for candidates of foreign nationality. Candidates with French nationality, including dual citizenship, are not eligible for this pathway.
- Age: Candidates must be under 25 years old on January 1, 2027.
- Language: Candidates must have sufficient proficiency in English and/or French to take the examinations. French proficiency is recommended because many courses in the Cycle Ingénieur are taught in French, but non-French-speaking admitted students may benefit from an intensive French language program.
- Academic background: Candidates must have a strong background in mathematics and in at least one scientific discipline among physics, computer science, chemistry, or life sciences.

Candidates must also be enrolled in the first cycle of university studies, such as a Bachelor's degree, at the time of the examinations. They must not have been enrolled in a Master's program in science or engineering, and they must have completed at least two years of higher education in science or engineering by the application deadline.

A candidate may apply a maximum of two times, provided that all eligibility requirements are met for each application.

1.2 Application Procedure and Required Documents

Applications are submitted online through the official application platform: Application platform

Candidates must complete the online form and submit academic and administrative documents.

Academic Documents

- Personal statement of 2 to 3 pages, presenting scientific interests, extracurricular interests, motivations, and future plans.
- Curriculum vitae, using the mandatory template.
- Course descriptions for all courses taken since the beginning of higher education, using the mandatory template.
- Complete official transcripts, validated by the president or dean of the home university.
- Official ranking certificates, including entrance examination ranking for university admission and end-of-year ranking for each academic year.
- Diplomas and degrees obtained, if applicable.
- Certificates for awards and distinctions mentioned in the application.
- Two letters of recommendation, written by professors or academic supervisors, using the mandatory template.
- Proof of French and/or English proficiency.

Administrative Documents

- Certificate of enrollment for the current academic year.
- Clear copy of a valid identity document, such as a national identity card or passport.
- Recent passport-style photograph.

Candidates should enter their surname and first name exactly as they appear on their identity document. Documents must be translated into French or English when necessary.

1.3 Examination Process

The FUI entrance exam has two main stages: *preselection* and *admission oral examinations*. The use of artificial intelligence during the examination process is strictly prohibited.

Preselection

Preselection is based on the evaluation of the application file. The admissions committee considers both academic and extracurricular elements.

Admission Oral Examinations

The admission stage consists of two scientific oral examinations.

Examination	Subject	Duration	Coefficient
Major Subject	Chosen among Mathematics, Physics, Computer Science, Chemistry, or Life Sciences	55 minutes	8
Minor Subject	Mathematics if the major is not Mathematics; otherwise chosen among Physics, Computer Science, Chemistry, or Life Sciences	55 minutes	5

Each oral examination includes scientific questions and a discussion about the candidate's motivation, scientific curiosity, general knowledge, and language level. The oral examinations are conducted remotely by videoconference, usually from an official examination center in the country of the candidate's home institution.

Scoring

The final scientific score is calculated using the following formula:

$$\text{Overall Scientific Score} = 0.75 \times \text{Scientific Oral Score} + 0.25 \times \text{Academic File Score}$$

Admission results are communicated after the jury's deliberation. Admitted candidates must confirm their acceptance through the applicant portal.

Contact

For questions about the international entrance exam: concours-international@polytechnique.fr

More detail information related to entrance exam: official website

Important note: This summary is based on the official entrance exam website. Since the document may be updated, candidates should regularly consult the official École Polytechnique website and application platform.

2. Mathematics

This section presents the recommended mathematics background for the École Polytechnique, FUI entrance exam.

2.1 Recommended Mathematics

Algebra

1. **Set Theory**
   - Operations on sets: union, intersection, complement, inclusion.
   - Characteristic functions of sets.
   - Maps and functions between sets.
   - Injective, surjective, and bijective maps.
   - Direct image and inverse image of a set.
   - Integer numbers, finite sets, and countability.
2. **Numbers and Algebraic Structures**
   - Composition laws and their basic properties.
   - Groups, rings, and fields.
   - Equivalence relations and quotient structures.
   - Real numbers and complex numbers.
   - Complex exponential and applications to plane geometry.
   - Polynomials and relations between roots and coefficients.
   - Elementary arithmetic in $\mathbb{Z}/n\mathbb{Z}$.
3. **Finite Dimensional Vector Spaces**
   - Free families, generating families, bases, and dimension.
   - Determinant of n vectors and characterization of bases.
   - Matrices and operations on matrices.
   - Determinant of a square matrix, rank, cofactors, and expansion along a row or column.
   - Linear maps and associated matrices.
   - Endomorphisms, trace, determinant, and rank.
   - Linear systems of equations.
4. **Reduction of Endomorphisms**
   - Stable subspaces.
   - Eigenvalues and eigenvectors of an endomorphism or a square matrix.
   - Similar matrices and geometric interpretation.
   - Characteristic polynomial.
   - Cayley-Hamilton theorem.
   - Reduction of endomorphisms in finite dimension.
   - Diagonalizable endomorphisms and matrices.
5. **Euclidean Spaces and Euclidean Geometry**
   - Scalar product and Cauchy-Schwarz inequality.
   - Norms and associated distances.
   - Finite dimensional Euclidean spaces.
   - Orthonormal bases and orthogonal projections.
   - Orthogonal group $O(E)$ and orthogonal symmetries.
   - Orthogonal matrices.

- Diagonalization of symmetric real matrices.
- Properties of orthogonal endomorphisms of $\mathbb{R}^2$ and $\mathbb{R}^3$.

Analysis and Differential Geometry

1. **Topology in Finite Dimensional Normed Spaces**
   - Open and closed sets.
   - Accumulation points and interior points.
   - Convergent sequences in normed vector spaces.
   - Continuous mappings.
   - Compact spaces and images of compact sets by continuous mappings.
   - Existence of extrema on compact sets.
   - Equivalence of norms in finite dimension.
2. **Functions of One Real or Complex Variable**
   - Derivative at a point and functions of class C^k.
   - Mean value theorem and Taylor's formula.
   - Primitives of continuous functions.
   - Usual functions: exponential, logarithm, trigonometric functions, and rational fractions.
   - Sequences and series of functions.
   - Simple convergence and uniform convergence.
3. **Integration on a Bounded Interval**
   - Integral of piecewise continuous functions.
   - Fundamental theorem of calculus.
   - Integration by parts.
   - Change of variable.
   - Integrals depending on a parameter.
   - Continuity under the integral sign.
   - Differentiation under the integral sign.
   - Cauchy-Schwarz inequality for integrals.
4. **Series and Power Series**
   - Series of real or complex numbers.
   - Simple convergence and absolute convergence.
   - Integral comparison criterion.
   - Product of absolutely convergent series.
   - Power series and radius of convergence.
   - Functions expandable in a power series on an interval.
   - Taylor series expansions of e^t, $\sin t$, $\cos t$, $\ln(1+t)$, and $(1+t)^a$.
5. **Differential Equations**
   - First order linear scalar differential equations.
   - Second order linear scalar differential equations.
   - Fundamental systems of solutions.
   - Linear systems with constant coefficients.
   - Method of variation of constants.
   - Basic notions on nonlinear differential equations.
6. **Functions of Several Real Variables**

- Partial derivatives.
- Differential of a function defined on $\mathbb{R}^k$.
- Chain rule.
- C^1-functions and Schwarz theorem for C^2-functions.
- Diffeomorphisms and inverse function theorem.
- Critical points, local extrema, and global extrema.
- Plane curves: tangent vector, arc length, and curvature.
- Surfaces in $\mathbb{R}^3$.
- Tangent plane to a surface defined by $F(x, y, z) = 0$.

2.2 Mathematics References

The references below will be completed later. Each reference can be a textbook chapter, lecture note, video lecture, online course, or problem set.

1. Set Theory: To be added
2. Numbers and Algebraic Structures: To be added
3. Finite Dimensional Vector Spaces: To be added
4. Reduction of Endomorphisms: To be added
5. Euclidean Spaces and Geometry: To be added
6. Topology in Finite Dimensional Normed Spaces: To be added
7. Functions of One Variable: To be added
8. Integration: To be added
9. Series and Power Series: To be added
10. Differential Equations: To be added
11. Functions of Several Variables and Differential Geometry: To be added

3. Physics

This section summarizes the recommended physics background for the École Polytechnique, FUI entrance exam.

3.1 Recommended Physic

General Tools for Physics

1. **Physical Constants and Orders of Magnitude**
   - Planck, Boltzmann, and Avogadro constants.
   - Electron charge and mass, speed of light in vacuum.
   - Electric permittivity and magnetic permeability of free space.
   - Orders of magnitude: Earth magnetic field, Earth radius, gravitational acceleration, visible wavelengths, atomic distances, Bohr radius, and nuclear size.
2. **Calculation Skills**
   - Common expansions near $x = 0$, such as $\sin x$, $\cos x$, e^x, $\ln(1 + x)$, and $(1 + x)^\alpha$.
   - Derivatives and primitives of elementary functions.
   - Product rule, quotient rule, integration by parts, and convergence of integrals.
   - Partial derivatives, total differential, and functions of several variables.
3. **Vector Calculus and Differential Operators**
   - Nabla operator ∇, gradient, divergence, curl, and Laplacian.
   - Circulation, flux, multiple integrals, and use of symmetry.
   - Stokes theorem and Gauss-Ostrogradski theorem.
4. **Differential Equations and Linear Algebra**
   - First order differential equations with separable variables.
   - Second order linear differential equations with constant coefficients.
   - Forced oscillations, resonance, and critical damping.
   - Wave equation, progressive monochromatic plane waves, wave vector, wavelength, frequency, and period.
   - Determinants, eigenvalues, eigenvectors, and diagonalization.
5. **Trigonometry and Fourier Series**
   - Basic trigonometric functions and common trigonometric identities.
   - Fourier series of sufficiently regular periodic functions.

Main Physics Chapters

1. **Mechanics**
   - Newton's laws, Galilean relativity, inertial and non-inertial reference frames.
   - Momentum, angular momentum, kinetic energy, potential energy, and conservation laws.
   - Central force motion, bound and scattering states, and Kepler's laws.
   - Motion in cylindrical and spherical coordinates.
   - Collisions, center of mass, and circular motion.
   - Rigid body rotation, moment of inertia, Koenig's theorem, and compound pendulum.
   - Fluid mechanics: velocity field, flow rate, mass balance, Euler equation, Bernoulli relation, pressure forces, and Archimedes' principle.
   - Applications: Lorentz force, charged particle motion, harmonic oscillations, damping, forced oscillations, and resonance.

2. **Electric Circuits**
   - Electric voltage, electric current, Kirchhoff's laws, Ohm's law, and superposition theorem.
   - Resistors, capacitors, coils, and their impedances in sinusoidal regime.
   - Charging and discharging of a capacitor.
   - Sinusoidal currents and voltages, maximum value, RMS value, and impedances in series or parallel.
   - Resonance in RLC circuits and connection with mechanical resonance.
3. **Electricity and Magnetism**
   - Electrostatics: Coulomb's law, electric field, Gauss theorem, electrostatic potential, Poisson equation, charge distributions, capacitors, electric dipoles, conductors, and dielectrics.
   - Magnetostatics: magnetic field, Biot-Savart law, Maxwell equations, Ampère's law, vector potential, magnetic dipoles, magnetic flux, electromagnetic induction, Faraday's law, and Lenz's rule.
   - Electromagnetic waves: Maxwell equations in vacuum, plane waves, frequency, wavelength, wave vector, phase velocity, polarization, energy density, Poynting vector, wave packets, group velocity, and waves in matter.
4. **Optics**
   - Geometrical optics: light rays, reflection, refraction, Snell-Descartes laws, total reflection, mirrors, lenses, conjugation, and magnification.
   - Wave optics: reflection and refraction of polarized plane waves, optical path, interference, Michelson interferometer, thin slabs, Fabry-Pérot cavity, diffraction, Huygens-Fresnel principle, rectangular slit, Young's slits, and diffraction gratings.
5. **Thermodynamics**
   - State functions: internal energy, entropy, enthalpy, free energy, free enthalpy, and their differentials.
   - Extensive and intensive variables, thermodynamic equilibrium, and heat capacities.
   - Perfect gas: monoatomic model, Maxwell-Boltzmann distribution, equipartition theorem, pressure and mean square velocity, barometric formula, real gases, and van der Waals gas.
   - First principle: internal energy, heat transfer, work, pressure work, enthalpy, Joule-Thomson expansion, and enthalpy of a perfect gas.
   - Second principle: entropy, entropy balance, reversible and irreversible processes, thermodynamic temperature, heat machines, Carnot theorem, phase equilibrium, Clapeyron formula, chemical potential, and Gibbs' phase rule.

3.2 Physics References

The following references will be added later. Each reference may be a textbook chapter, lecture note, video, or online course.

1. Physical Constants and Orders of Magnitude: To be added
2. Mathematical Tools for Physics: To be added
3. Vector Calculus and Differential Operators: To be added
4. Differential Equations and Linear Algebra: To be added
5. Trigonometry and Fourier Series: To be added
6. Mechanics: To be added

7. Electric Circuits: To be added
8. Electrostatics: To be added
9. Magnetostatics: To be added
10. Electromagnetic Waves: To be added
11. Geometrical Optics: To be added
12. Wave Optics: To be added
13. Thermodynamics: To be added